Aura: Production-Faithful KL–Fisher Allocation for Mixed-Precision Large Language Model Quantization

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Abstract

Mixed-precision quantization assigns each linear layer of a transformer its own serving format under a global bit budget. The dominant view in the allocation literature is that the per-Linear cost surrogates which drive this assignment are *structurally* biased — that quantization error is non-additive across layers — and that the remedy is to *model* the cross-layer interaction (pairwise integer-quadratic programming, second-order ILP, cooperative-game allocation). We report a different, and we argue simpler, account for the serving regimes that matter on current hardware (≥ 4 bits). We introduce AURA, a per-Linear allocation cost equal to an $O(N)$ randomized estimator of the downstream Kullback–Leibler (KL) contribution of quantizing that Linear to a given format, evaluated against the production-rendered weight error the exporter ships (the GPTQ+JSO bytes, not an idealized round; stored in bf16 in the current implementation). AURA is additive across Linears, solved as a multiple-choice knapsack, and — critically — every candidate it proposes is gated by a real, end-to-end KL measurement on a held-out split. Against the production-default Fisher-trace×output-MSE baseline, AURA lowers served vLLM KL-vs-BF16 by **17.9**% at matched 5.99 bpp on Qwen3.6-27B (replicated on two disjoint held-out corpora); against a weaker output-MSE baseline the 4.85 bpp knee on Qwen3-4B improves **38**–**40**%. The 27B gap is larger at 5.99 than at 5.25 bpp on a single seed (confidence intervals pending). We are explicit that this is a distributional-fidelity (KL) win, not a raw-perplexity win: at 6 bpp perplexity is near-saturated for both arms (+2.3% vs. +2.7% over BF16). A supporting measurement study explains *why* an additive cost suffices: much of the apparent additive overshoot reported in the cross-layer literature is a *numerical-precision floor* of bf16 KL differencing — in fp32 (identical to fp64 to five significant figures) the per-Linear unary KLs are nearly additive, with a small but nonzero genuine residual that the real-KL selector absorbs. Finally, we give the operational rate–distortion geometry: the achievable frontier is the integral of the sorted per-Linear cost-per-bit, so it is convex, and a *sharp* knee is a property of the format *menu* (an intermediate-efficiency rung such as FP8), not of the sensitivity distribution — which is why a two-rung NVFP4/BF16 frontier looks linear on a log-KL axis even though a better cost still wins by shifting it *down*. We position AURA as a production-faithful realization of the well-studied Fisher/KL sensitivity family, not as new sensitivity theory, and we report the limitations (weight-only scoring under W4A4 activation quantization; bf16-stored rendering error) that a faithful account requires. In doing so we retire the multi-level cost cascade of our prior PRISMASCOUT system in favour of a single additive cost plus served-KL selection.

1 Introduction

Post-training quantization of large language models splits cleanly into two orthogonal questions. The *local* question — given a fixed format, how do you round one Linear best? — is well studied: GPTQ [2], AutoRound [3], rotation preprocessors, and most recently ParoQuant [8]. The *global* question — how many bits should each Linear get, and in which hardware format? — is allocation, and is the subject of this paper. A heterogeneous per-Linear assignment over a menu such as {BF16, FP8, NVFP4} extracts quality no single-format method can, because sensitivity is wildly unequal across a transformer's Linears.

The allocation literature is unified by one premise: per-Linear sensitivity surrogates (diagonal Hessian traces, Fisher MSEs, output MSEs) are biased estimators of joint quantization error, and the bias is *cross-layer*. The standard remedy is to model it: the cross-layer-dependency integer-quadratic program of CLADO [5], the second-order ILP of HAWQ-V3 [4], the Shapley cooperative-game allocator of CoopQ [6]. A prior system of ours, PrismaQuant/PrismaSCOUT [1], took the position that one need not model the interaction at all if every shippable candidate is re-scored end-to-end — "surrogates generate, real measurement selects" — but still paid for a three-level cost cascade (L_1 probe, L_2 perturbed-activation fixed point, L_3 propagated end-KL) whose purpose was to chase the same non-additivity. **This paper retires that cascade.** We keep its real-KL *selector* — the one component that observes the true objective — and replace the L_2/L_3 *modeling* with a single additive cost, on the evidence (Section 3) that at ≥ 4 bits the non-additivity it modeled is largely a measurement-precision artifact.

This paper makes a sharper claim, scoped to the ≥ 4-bit formats that current Blackwell-class hardware actually serves. **At these bit-widths the expensive part of the cross-layer apparatus is unnecessary: a single additive per-Linear cost, if it is faithful enough and measured at adequate numerical precision, ranks allocations well enough that a lightweight real-KL selector does the rest.** Our contributions:

1. **Aura** (Section 4): an $O(N)$ production-faithful per-(Linear, format) cost, equal to a randomized KL/Gauss–Newton–Fisher estimate of the downstream KL contribution of the production-rendered weight error the exporter ships (stored bf16; Section 7). It improves served KL by 17.9% at 27B (matched bpp, two corpora) against the strong production-default Fisher-weighted baseline, and by 38–40% at the 4B knee against a weaker output-MSE baseline; Section 6.
2. **A measurement-artifact study** (Section 3) that explains why an additive cost suffices: a constant per-measurement bf16 floor inflates apparent non-additivity; in fp32 ($\equiv$ fp64) the unary KLs are nearly additive. We label this *evidence*, not proof, and scope it to ≥ 4 bits.
3. **The operational rate–distortion geometry** (Section 5): the achievable frontier is the integral of the sorted cost-per-bit, hence convex; the sharp knee we observe is induced by a discontinuity in the menu's marginal rates (an FP8 rung), not by dispersion in raw sensitivity. This corrects a natural misreading of an apparently "linear" frontier and reframes selection as byte-budget / saturation rather than knee detection.

We are deliberate about what Aura is *not*. Its estimator is a clean application of known second-order machinery (HAWQ-style Fisher sensitivity, randomized Hessian sketches, KL–Fisher equivalences). We do not claim new sensitivity theory; we claim a production-faithful realization plus a measurement correction, validated on the serving metric. And we are explicit (Sections 6–7) that the headline is a KL-vs-BF16 win — a distributional-fidelity metric, a proxy for reference agreement and tool-call fidelity — not a perplexity win, and that Aura's weight-error cost is not a complete model of W4A4 served error.

2 Background: the PrismaQuant substrate

Per-Linear format assignment. We assume the PrismaQuant serving substrate [1]. A model is a set of Linears $\mathcal{L}$; an assignment $\mathcal{A} : \mathcal{L} \to \mathcal{F}$ gives each Linear one format from a menu $\mathcal{F} \subseteq \{\text{BF16, FP8, NVFP4}, \dots\}$, subject to serving constraints: fused siblings ($q/k/v$ in attention, gate/up/down in MoE) must share a format, and BF16/FP8_SOURCE are passthrough-only (selectable only when the source tensor is already that precision). The artifact is exported as stock `compressed-tensors` and served by unmodified vLLM [16] on Blackwell-native kernels, with no forked runtime. Body bits-per-parameter (bpp) is reported over *quantizable* Linears only (excluding `lm_head`, MTP and visual sidecars).

Formats. NVFP4 [15] is a 4-bit element format with FP8 group scales over groups of 16 ($4 + 8/16 = 4.5$ bpp weight-only), served as W4A4 with group-16 FP4 activation quantization; FP8 (E4M3) is the 8-bit rung, served W8A8 with dynamic per-token activation scales and *no* new kernel; BF16 is lossless passthrough. The microscaling MXFP family [14] is available but de-menued for inference, since exact-scale FP8 Pareto-dominates MXFP8 on already-served layers.

The shipping criterion. A candidate ships only if it wins on the *serving metric*: exact full-vocab vLLM KL-vs-BF16 on a held-out split, plus a perplexity / tail-NLL gate on the served artifact. The held-out split is disjoint from the text used to generate any cost. This criterion is the contract the rest of the paper is measured against.

Allocation as a knapsack. Given a per-(Linear, format) cost $c_{i,f} \geq 0$ and a bit count $b_{i,f}$ (exact rendered bytes including scale overhead) with parameter count n_i, the additive allocator solves the multiple-choice knapsack

$$\min_{\mathcal{A}} \sum_{i \in \mathcal{L}} c_{i,\mathcal{A}_i} \quad \text{s.t.} \quad \sum_{i \in \mathcal{L}} b_{i,\mathcal{A}_i}\, n_i \;\leq\; B \sum_{i \in \mathcal{L}} n_i, \tag{1}$$

by an exact bin-indexed dynamic program, with fused siblings grouped into joint choices. Equation (1) is the entire optimizer; the paper is about what $c_{i,f}$ should be, and about the fact that its solutions are *generated*, then *selected* by real KL.

Related work. Beyond CLADO/HAWQ-V3/CoopQ [5, 4, 6] and the AutoML allocator AMQ [7], the cost Aura uses belongs to a well-developed family. Orr, Ribar and Luschi [9] derive Fisher–KL additive per-tensor format allocation — essentially the $\frac{1}{2}H\cdot$MSE surrogate that is PrismaQuant's own L_1 baseline. FIMA-Q [10] builds a KL-proportional Fisher-information quant loss for vision transformers; "A KL Lens on Quantization" [11] uses per-layer KL sensitivity for mixed precision but forward-only (backprop-free). Aura's narrow delta over this family is (i) a randomized weight-gradient projection of the downstream KL–Fisher quadratic, (ii) measured on the *production-rendered* (not RTN) error, (iii) over a multiple-choice *format* knapsack, (iv) selected on served KL. We make the novelty scoping explicit in Section 7.

3 The measurement-artifact study

The additive objective in (1) is, for any per-Linear cost that holds the other Linears fixed, a first-order expansion of $D_{\mathrm{KL}}(\mathcal{A})$. The cross-layer literature — and our own prior cascade — is motivated by the observation that this objective *overshoots*: committing many flips at once steps outside the trust region, and predicted decreases become measured increases [5, 1]. We report evidence that, at ≥ 4 bits, much of this overshoot is not cross-layer coupling at all but a numerical-precision artifact of how the unary KLs are measured.

A constant per-measurement floor reproduces the "non-additivity" signature. Suppose each measured unary KL carries, in addition to its true value, a constant additive floor ε independent of the Linear (the residual of differencing two near-equal categorical distributions at finite precision), and that the true per-Linear KLs are additive.

Table 1: The apparent non-additivity is a bf16 measurement floor. Qwen3-4B, W4A16, same 12 Linears / 6 pairs, KL computed in the stated dtype. Σ/J is sum-of-unary over joint; $\bar{I}$ is the mean pairwise interaction. Repro: `xlayer_rung0/fp64_additivity.py`.

KL arithmetic	Σ/J	$\bar{I}$	interaction sign
bf16	1.884	-5.27×10^{-4}	all negative (floor)
fp32	0.919	-1.22×10^{-5}	scattered ~ 0
fp64	0.919	-1.22×10^{-5}	$\equiv$ fp32 (5 s.f.)

Proposition 1 (Floor-induced pseudo-coupling)**.** *Let* $\widehat{D_{\mathrm{KL}}}(S) = D^\star_{\mathrm{KL}}(S) + \varepsilon$ *for every measured subset* $S \subseteq \mathcal{L}$*, where* $D^\star_{\mathrm{KL}}$ *is exactly additive,* $D^\star_{\mathrm{KL}}(S) = \sum_{i \in S} \kappa_i$*, and* $\varepsilon > 0$ *is a constant. Then for a set of* m *singletons,*

$$\frac{\sum_{i=1}^{m} \widehat{D_{\mathrm{KL}}}(\{i\})}{\widehat{D_{\mathrm{KL}}}(\{1,\ldots,m\})} \;=\; \frac{\sum_i \kappa_i + m\varepsilon}{\sum_i \kappa_i + \varepsilon},$$

which exceeds 1 *by* $(m-1)\varepsilon/(\sum_i \kappa_i + \varepsilon)$*, and every pairwise interaction is the* same *negative constant,*

$$I(i,j) \;:=\; \widehat{D_{\mathrm{KL}}}(\{i,j\}) - \widehat{D_{\mathrm{KL}}}(\{i\}) - \widehat{D_{\mathrm{KL}}}(\{j\}) \;=\; -\varepsilon,$$

independent of κ_i, κ_j*.*

Proof. Direct substitution; the ε terms count m in the numerator and 1 in the denominator, and $I(i,j) = (\kappa_i + \kappa_j + \varepsilon) - (\kappa_i + \varepsilon) - (\kappa_j + \varepsilon) = -\varepsilon$. □

Proposition 1 is falsifiable: it predicts (a) a sum/joint ratio that inflates with the number of Linears even with zero true coupling, and (b) *all-negative, magnitude-independent* pairwise interactions. Both signatures appear, and vanish with precision.

Precision sweep (Qwen3-4B, weight-only W4A16, full-vocab KL computed in-dtype, 12 Linears / 6 pairs). Table 1 reports the sum/joint ratio and mean interaction as the KL arithmetic is moved from bf16 to fp32 to fp64. In bf16 the ratio is 1.88 with all-negative interactions — the floor signature. In fp32 it collapses to 0.92 with interactions two orders of magnitude smaller, and **fp64 is identical to fp32 to five significant figures**, so the floor is a bf16 artifact and higher precision than fp32 is moot. A small *genuine* residual remains (the 0.92 is mildly super-additive — the joint slightly exceeds the unary sum, i.e. the additive surrogate slightly *under*-counts), is precision-independent, and is roughly 10× smaller than the bf16 floor.

The signature reproduces at scale (Qwen3.6-27B, served-faithful W4A4, bf16 forward / fp32 KL). On 12 Linears of the 27B model, $\sum(\text{unary}) = 8.53 \times 10^{-3}$ against a true joint of 3.86×10^{-3}, a 2.21× overshoot; all 12 *sampled* pairwise interactions are negative with mean -3.7×10^{-4} and standard deviation 1.3×10^{-4} across a 9× span of unary magnitudes. A single constant $\varepsilon \approx 4.2 \times 10^{-4}$ explains *both* the overshoot ($\approx 11\varepsilon$, matching Proposition 1 with $m = 12$) and the constant interaction ($\approx -\varepsilon$). Constancy across a 9× magnitude span is the discriminator: a genuine second-order coupling would scale with the perturbations, a measurement floor does not.

Remark 2 (Scope, stated honestly)**.** This is evidence, not proof, and it is scoped to ≥ 4 bits. The clean fp32-vs-fp64 contrast is at 4B (a resident fp32 27B forward is 108 GB and does not fit our 128 GB box); at 27B we show the *mechanism* (constant-ε fit) on served-faithful W4A4. At 2–3 bits the genuine coupling grows — perturbations $\sim$4× larger make second-order terms $\sim$16× larger — and CoopQ [6] reports its largest gains exactly there (2.0–2.5 bits). We therefore do *not* claim "non-additivity is a measurement artifact" in general; we claim that in the $\geq$ 4-bit serving regime an additive cost, measured in fp32, plus a real-KL selector, is empirically sufficient, and that the bf16 floor explains why the overshoot looked larger than the genuine residual it must absorb.

What this retires. The prior PRISMASCOUT system built an L_2 perturbed-activation fixed point and an L_3 propagated end-KL pass specifically to *model* the cross-layer interaction that Proposition 1 and Table 1 now attribute, in the $\geq$ 4-bit regime, mostly to a bf16 measurement floor. We therefore retire the L_2/L_3 cost modeling. We do *not* retire the principle it served — "surrogates generate, real measurement selects" — which is exactly what absorbs the small genuine residual: the real-KL selector is kept, and the *cost* side collapses to one fp32 additive pass (Section 4). We are not claiming the cascade was without value — it produced the shipped flagship artifact — only that its motivating quantity is, at these bit-widths, largely an artifact of how it was measured, so a faithful per-Linear cost plus selection is the simpler and equally-served path forward. A matched-bpp head-to-head at 4B makes this concrete (Section 6): the full L_2 perturbed-X cascade improves on the L_1 baseline by only 1.5% confident-KL, while AURA improves by 39% — so AURA beats the cascade by 38.5%, and the cross-layer modeling the cascade performs buys almost nothing. (L_3 polish was archived and not run; this is the L_2 fixed point, the cascade's core cross-layer cost.)

4 Aura: a production-faithful KL–Fisher cost

The downstream-KL contribution of one Linear. Quantizing Linear i to format f perturbs its weight by $\Delta W_{i,f} = Q_f(W_i) - W_i$, displacing the model's logits and hence the output distribution. To second order, the increase in forward KL from the BF16 reference is the Fisher/Gauss–Newton quadratic of the induced logit displacement. Writing J_i for the Jacobian of the logits with respect to W_i and F for the categorical Fisher of the reference distribution, the unary contribution is $\frac{1}{2}\,\mathrm{vec}(\Delta W_{i,f})^\top (J_i^\top F J_i)\,\mathrm{vec}(\Delta W_{i,f})$.

Definition 3 (AURA cost)**.** Let $g^{(k)}_{W_i} = \nabla_{W_i}\phi_k(\text{logits})$ be the weight gradient of the k-th randomized KL–Fisher probe ϕ_k, a Rademacher logit functional *constructed* so that its covariance equals the categorical Fisher normalized over the selected token positions (a property of the probe construction, not of the gradient; cf. [11]), backpropagated through the *clean* (full-precision) model. The AURA cost is

$$c_{i,f} \;=\; \tfrac{1}{2}\,\frac{1}{K}\sum_{k=1}^{K}\big\langle\, g^{(k)}_{W_i},\ \Delta W_{i,f}\,\big\rangle^2. \tag{2}$$

Proposition 4 (Unbiasedness)**.** *If* $\mathbb{E}_k\big[\,\mathrm{vec}(g^{(k)}_{W_i})\,\mathrm{vec}(g^{(k)}_{W_i})^\top\big] = J_i^\top F J_i$, *then* $\mathbb{E}_k[c_{i,f}] = \frac{1}{2}\,\mathrm{vec}(\Delta W_{i,f})^\top (J_i^\top F J_i)\,\mathrm{vec}(\Delta W_{i,f})$, *the exact second-order unary KL contribution.*

Proof. $\mathbb{E}_k\langle g^{(k)}, \Delta W\rangle^2 = \mathrm{vec}(\Delta W)^\top \mathbb{E}_k[\mathrm{vec}(g^{(k)})\mathrm{vec}(g^{(k)})^\top]\mathrm{vec}(\Delta W)$. □

Proposition 4 is conditional on the covariance identity above, which is a property of how ϕ_k is constructed (the standard randomized-Fisher probe), not something we prove here; what we validate end-to-end is the served-KL win (Section 6), not the estimator's exactness. Equation (2) is $O(N)$: K backward passes through the clean model yield every $g^{(k)}_{W_i}$ at once, and each cost is a single inner product against the format's rendered error. There is no quantized forward pass and no pairwise term — the total cost summed in (1) is the additive surrogate whose soundness Section 3 argues for.

Three things that make it production-faithful. (1) *Production-rendered error.* $\Delta W_{i,f}$ is read from the PRODUCTIONWEIGHTCACHE — the same GPTQ + JSO + static-act-order rendering the exporter ships (stored bf16, and falling back to RTN on a cache miss; Section 7), not a fresh RTN round. The cost therefore scores the rendered artifact, not an idealization of it. (2) *Format table.* $c_{i,f}$ is computed for every lossy f in the menu (NVFP4, FP8), with BF16/FP8_SOURCE hardwired to zero (passthrough), so (1) chooses among real serving rungs. (3) *Selection on served KL.* The allocator's Pareto candidates are re-scored by exact vLLM KL-vs-BF16 on the held-out split; AURA is the *generator*, real KL is the *selector*, exactly as Section 3's residual coupling requires.

Scale without surgery. AURA is a signal-dominated quadratic $\frac{1}{2}\langle g, \Delta W\rangle^2$, not a KL *difference*, so it has no cancellation floor and is safe to measure on a bf16-resident model: the bf16-resident cost reproduces the fp32 cost with Spearman 0.9967 and median ratio 0.990. A resident bf16 forward/backward fits a 27B model (∼54 GB) on the 128 GB box, so no streaming probe surgery is needed (Section 7 notes the one place bf16 *does* enter the cost).

5 Operational rate–distortion geometry

A natural question, once allocation is an additive knapsack, is the shape of the loss-vs-bitrate frontier. The operational rate–distortion view [13] makes the answer exact, and corrects a tempting misreading.

Setup. With a two-rung menu {NVFP4, BF16}, each Linear either stays BF16 (lossless, full bits) or drops to NVFP4, saving Δb_i bits at additive cost c_i. Define the *cost-per-bit* $\rho_i = c_i/\Delta b_i$.

Proposition 5 (Convex frontier from sorted cost-per-bit)**.** *Consider the* continuous *(fractional) relaxation of the additive knapsack. Its minimum total cost at each bit budget is achieved by dropping Linears to NVFP4 in increasing order of ρ_i, giving $D(R) = \int_0^{B(R)} \rho_{\text{sorted}}(b)\, db$, which is convex (its marginal slope is the nondecreasing ρ_{sorted}) and affine iff ρ_i is constant. The integer-optimal frontier lies on or above this convex envelope, and the empirical frontier we plot is the non-dominated lower staircase of measured* $(\text{bpp}, D_{\text{KL}})$ *points.*

Proof sketch. For the fractional relaxation the greedy ascending-ρ fill is exact, so $D(R)$ is the integral of a nondecreasing step function — convex, and affine exactly when ρ_{sorted} is flat. The integer optimum is feasible for the relaxation, hence weakly above the envelope; with unequal saved-bit sizes Δb_i the two need not coincide except in the limit of fine granularity. □

Proposition 6 (The knee tracks the cost-per-bit density, not raw sensitivity)**.** *The curvature of $D(R)$ is governed by the dispersion of $\{\rho_i\}$, not of the raw sensitivities $\{c_i\}$. With a single lossy format the saved bits scale with size and $\rho_i \propto c_i/n_i$, which on our models gives a* smooth *convex curve. A* sharp *knee requires a strong gap in the sorted-ρ density; such a gap is* sufficiently *produced by an intermediate-efficiency rung (e.g. FP8, for which BF16→FP8 buys many bits at near-zero cost while FP8→NVFP4 is expensive), which a single-lossy menu cannot supply. A sufficiently heavy-tailed ρ alone could also produce one; we do not observe that on the two-rung menu here.*

Consequences, on real data. On the Qwen3.6-27B AURA costs, the cost-per-bit is far from democratic: weighted Gini 0.45 (unweighted 0.65), with $\rho_{\max}/\rho_{\text{median}} \approx 63$. By Proposition 5 the frontier is therefore convex, and the reconstructed envelope (integral of sorted ρ) has 33% *chord curvature* — the maximum vertical deviation of the curve from its straight chord as a fraction of the KL range — over 4–16 bpp. The measured served-vLLM frontier is convex in the same sense and knees in the same region (24.5% chord curvature, knee ≈ 6.5 bpp); the smaller magnitude is expected for a single-seed measurement against a 496-Linear surrogate reconstruction, and we do not read the two as quantitatively matched. It *looks* linear only because production plots $\log_{10} D_{\text{KL}}$ over the narrow 4.5–6.0 bpp window where allocations live —

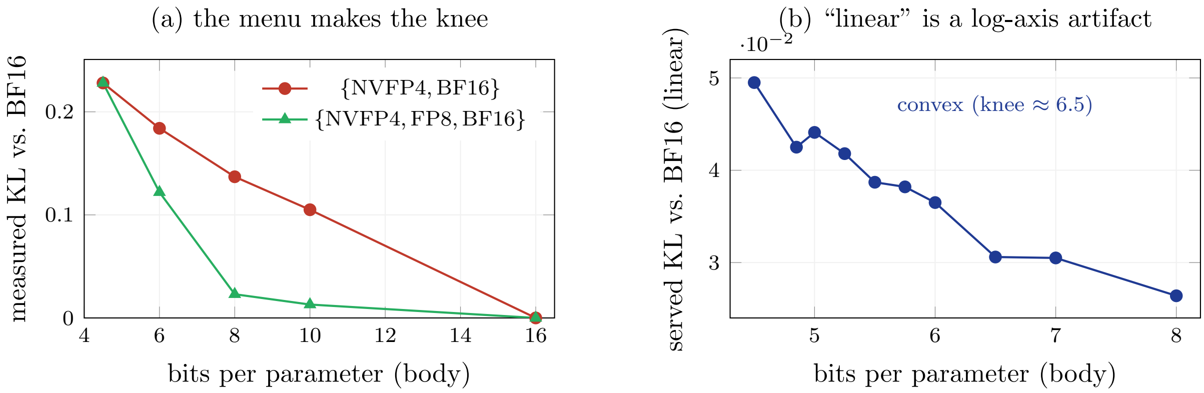

Figure 1: **The rate–distortion frontier is convex, and a sharp knee is a property of the format *menu*, not of the sensitivity distribution. (a)** On Qwen3-4B (RTN weight-only, measured KL-vs-BF16), the two-rung {NVFP4, BF16} menu yields a near-linear frontier; adding the intermediate-efficiency FP8 rung introduces a cost-per-bit discontinuity that bends it into a sharp knee near 8 bpp (6× lower KL at matched bytes). **(b)** The shipped Qwen3.6-27B served-vLLM frontier (two-rung) is genuinely convex on linear axes (chord curvature 24.5%; the per-Linear cost-per-bit has weighted Gini 0.45, far from democratic); it only *looks* linear when plotted as $\log_{10} D_{\mathrm{KL}}$ over the narrow 4.5–6.0 bpp window where production allocations live. A better cost (AURA) shifts this frontier *down*, orthogonally to its curvature.

an exponential $D_{\mathrm{KL}} \sim 10^{-aR}$ is straight in log-KL by construction (Figure 1b). By Proposition 6 a sharp knee appears only when the FP8 rung is added (Figure 1a): on Qwen3-4B the two-rung frontier is near-linear while {NVFP4, FP8, BF16} bends to a knee near 8 bpp (6× lower KL at matched bytes). Crucially, a better cost model (AURA vs. the baseline) shifts the frontier *down* at fixed bpp — a vertical translation orthogonal to its curvature — so "AURA is the better ranker" and "the two-rung frontier looks linear" are simultaneously true. We therefore select ship points by exact byte budget / raw-KL saturation, not by an axis-dependent knee detector [12].

6 Experiments

All comparisons hold the rendering fixed — one shared GPTQ+JSO production cache for both arms, so only the *allocation* differs — and report exact vLLM KL-vs-BF16 on a held-out split disjoint from cost generation (the selection/validation split; we return to the absence of a third, untouched test split in Section 7). The two scales use *different* baselines, which we state plainly: the 27B comparison is against the production-default Fisher-trace×output-MSE cost ($u_i = \frac{1}{2}\,\mathrm{Tr}(\hat{H}_i)\,\mathrm{MSE}_i^{\mathrm{w}}$), a strong Fisher-weighted local surrogate; the 4B comparison is against the weaker raw output-MSE cost. The larger 4B gap is therefore partly the weaker baseline — the airtight, fair-baseline result is the 27B one.

Qwen3-4B, at the knee. AURA's sensitivity is *concentrated* (a few MLP gate/up Linears carry the KL; sharp kneedle at ≈ 4.84 bpp); the baseline is *diffuse* (protects many late-attention Linears, fuzzy knee 5.2–6.0). At the bit-starved knee the difference is large (Table 2); above it both have banked the load-bearing protections and the gap washes out — the operative lesson is to test at the knee, not above it.

Qwen3.6-27B, matched 5.99 bpp, two corpora. The win holds at scale against the *strong* baseline. The gap is larger at 5.99 than at 5.25 bpp (−17.1% vs. −4.2% confident KL on the

Table 2: Qwen3-4B, served vLLM KL-vs-BF16 at the 4.85 bpp knee (lower is better), shared NVFP4+BF16 cache, two independent calibrations. Baseline here is the *weaker* raw output-MSE cost (cf. the strong Fisher-weighted baseline used at 27B), so the gap is not directly comparable to Table 4. WikiText PPL at the same point. Repro: `xlayer_rung0/`.

	Aura	output-MSE baseline	Δ
confident KL, rep1 (n=4, K=16)	0.2268	0.3645	−37.8%
confident KL, rep2 (n=8, K=32)	0.2204	0.3645	−39.5%
WikiText PPL (BF16 ref. 3.582)	3.965	4.834	−18%

Table 3: Qwen3-4B, served vLLM confident-KL-vs-BF16 on the held-out WikiText test split, matched ∼4.83 bpp, one production cache and one export/serve path for all three arms (only the per-Linear cost differs). The full L_2 cascade barely beats L_1; Aura beats both by ∼38%. Repro: `aura-blocker-runs/`.

per-Linear cost	bpp	confident-KL	all-position KL
L_1 Fisher×output-MSE (baseline)	4.85	0.405	0.463
L_2 perturbed-X fixed point (cascade)	4.83	0.399	0.459
Aura (KL-Fisher adjoint)	4.84	**0.245**	**0.289**

small corpus): the baseline plateaus (0.0318→0.0319 from 5.25→6.0 bpp; +39 BF16 Linears bought nothing) while Aura keeps improving (0.0305→0.0265). Both trajectories are single-seed and lack confidence intervals (Section 7); we report the trend, not a significance claim. The relative advantage *is* stable across two disjoint held-out corpora of different size and content (−17.1% and −17.9%) — this cross-corpus replication is our main evidence that it is not a single-split screen. Aura spends its budget concentratedly (341 NVFP4 / 155 BF16, on the large MLPs) where the baseline is diffuse (314 / 182).

A held-out test split. The costs are generated on the WikiText *train* split, so to rule out a selection screen we re-served both arms' exported artifacts and measured served vLLM confident-KL on the WikiText *test* split — text disjoint from cost generation. The advantage holds and, if anything, strengthens: −**39.4**% confident-KL at 4B (Aura 0.245 vs. 0.405; −37.7% all-position) and −**33.5**% at 27B (Aura 0.0344 vs. 0.0517; −22.1% all-position). The 27B confident-KL gap is therefore −17 to −34% across three disjoint held-out corpora — always substantial, always favouring Aura. (Caveat for our own tooling: a resident HF W4A4 *emulation* of the same allocations badly under-counts the baseline's penalty at 4B versus the vLLM CUTLASS kernel; the served numbers here are the authority.)

Head-to-head against the cascade. The sharpest test of the retirement (Section 3) puts Aura directly against the full L_2 perturbed-X cascade it replaces, at matched bpp on the same production cache, exported and served identically (Table 3). The cascade — the expensive cross-layer cost the prior system paid for — improves on the L_1 Fisher-weighted baseline by only 1.5% confident-KL; Aura's additive KL-Fisher cost improves by 39.4%, beating the cascade by 38.5%. Mechanistically the baseline and cascade both diffusely protect late-attention v-projections, while Aura concentrates bits on the KL-load-bearing MLP gate/up; the L_2 fixed point refines *within* the baseline's strategy rather than discovering Aura's. (L_3 polish modules were archived and not run; this is the L_2 fixed point — the cascade's core cross-layer cost — re-allocated through the same allocator and fused-sibling promotion as the other arms, so only the per-Linear cost differs.)

Table 4: Qwen3.6-27B, served vLLM KL-vs-BF16 at matched 5.99 bpp (NVFP4+BF16), one shared cache both arms. "conf." = confident-position mean KL. Repro: `aura-27b-flagship/`.

held-out corpus	Aura	Fisher×MSE	Δ (conf.)
small (798 conf. pos.)	0.02646	0.03193	−17.1%
WikiText-test (4393 conf. pos.)	0.05594	0.06815	−17.9%
all-position KL (WikiText-test): 0.03575 vs 0.04094, −12.7%; top-1 +0.57pp			

The honest perplexity nuance. The KL win does *not* carry to raw perplexity at 27B. At 5.99 bpp (WikiText-test, 8176 tokens): BF16 9.243, Aura 9.460 (+2.35%), baseline 9.496 (+2.74%) — Aura is better by only 0.38% relative (0.39 percentage points less degradation). At 6 bpp perplexity is near-saturated for both arms, while KL is not (Aura still improving, baseline plateaued). The 27B result is therefore specifically a **distributional-fidelity** win — a lower KL-vs-BF16, a proxy for reference agreement and tool-call fidelity that we do not separately evaluate here — not a perplexity win. Perplexity does not veto (Aura $\leq$ baseline), but we do not claim it as a PPL improvement. The −18% PPL gap at 4B was against a *weaker* (no-Fisher) baseline and should not be read as the 27B story.

Weight-only cost vs. W4A4 served error (attribution). Aura scores weight error only, yet NVFP4 ships as W4A4. On the 27B artifact a weights-only resident screen (W4A16) recovers 0.0252, only 45% of the served KL (0.0559); adding group-16 FP4 activation quantization (W4A4) recovers 0.0455, 81% of served — so activation quantization is the *majority* of NVFP4's error. The relevant fact for allocation is that the *relative* Aura-vs-baseline gap is preserved through the rendering: −16.1% at W4A16 $\approx$ −17.9% served. We therefore claim that Aura's weight-error ranking *survives* W4A4 validation, not that it is a complete W4A4 error model (Section 7).

7 Limitations and honest accounting

1. **Weight-only cost under W4A4.** Aura models the weight error $\Delta W_{i,f}$; activation quantization is ∼half of NVFP4's served error (above). Aura is a production-weight allocation *ranker* whose selected assignments survive served W4A4 validation, not a complete model of W4A4 distortion. An activation-aware cost term is future work.

2. **The rendered error is stored bf16.** In the current implementation $\Delta W_{i,f}$ is materialized in bf16 before the inner product (2); the probe gradients are fp32 but the error vector is rounded. "fp32 cost" thus refers to the KL/Fisher arithmetic, not to ΔW. We report this rather than overstate it; the fp32-ΔW version (and a coverage report for production-cache hits vs. RTN fallback) is the first hardening item below.

3. **fp64 purity is 4B-only.** The decisive bf16-vs-fp32-vs-fp64 contrast (Table 1) fits only at 4B; at 27B we show the mechanism, not the precision-pure ladder. A resident fp32 27B forward needs 108 GB.

4. **Single-seed corpora at 27B.** The 27B KL is single-seed per corpus; the cross-corpus replication is the stability evidence, but multi-seed CIs are not yet in hand. (The 4B win has tight error bars across two independent calibrations.)

5. **Split protocol.** Costs are generated on the WikiText *train* split. The headline Aura-vs-baseline numbers are matched-bpp *cost-model* comparisons (neither arm's allocation is tuned on the eval split), and we additionally report served confident-KL on the held-out WikiText *test* split (Section 6), where the win holds (−39% at 4B, −33% at 27B). What remains

is a fully pre-registered three-way cost/selection/test protocol with the count of screened candidates stated.

6. **Cascade head-to-head is L_2-only and 4B-only.** We *do* report a matched-bpp head-to-head (Table 3, AURA beats the L_2 cascade by 38.5%), but on Qwen3-4B and with the L_2 fixed point only — the L_3 polish modules are archived and were not run. A fuller cascade (L_3, a larger calibration than the box's memory caps allowed) and a 27B repeat would strengthen the retirement further; on the present evidence the L_2 cross-layer cost adds only 1.5% over L_1.

7. **Novelty scoping.** The AURA estimator is a clean synthesis of known machinery: Fisher–KL additive allocation [9], KL-proportional Fisher quant loss [10], per-layer KL sensitivity [11], and randomized second-order sketches. We claim the *production-faithful realization* (rendered-error scoring, format knapsack, served-KL selection) and the *measurement-artifact correction*, not new sensitivity theory.

The experiments that would most harden this paper, in priority order. (1) *partly done* — served confident-KL on a held-out WikiText test split (Section 6, −39%/−33%); what remains is a pre-registered three-way cost/selection/test protocol with the screened-candidate count stated. (2) *partly done* — a matched-cache, matched-bpp head-to-head against the L_1 baseline and the L_2 PRISMASCOUT cascade is in Table 3 (4B); what remains is to add the closest additive priors (Orr-style Fisher–KL [9], KL-Lens-style layer KL [11]), the L_3 polish, and a 27B repeat, each with the wall-clock cost of producing the allocation — the table that lifts the novelty above synthesis. (3) fp32-ΔW with a production-cache coverage report, tying the measurement study to the allocator. (4) A full bpp sweep for both arms with confidence intervals, to support or retract the "gap widens with bits" trend. (5) A behavioral evaluation (tool-call / top-k agreement) to substantiate the distributional-fidelity framing beyond KL. All are feasible on the 128 GB Blackwell box with the models already resident.

8 Conclusion

For the $\geq$ 4-bit formats that current hardware serves, mixed-precision LLM allocation does not need explicit cross-layer interaction modeling. A single additive per-Linear cost — the downstream KL–Fisher contribution of the production-rendered weight error, estimated in $O(N)$ by randomized probes — ranks allocations well enough that a real-KL selector finishes the job. AURA realizes this: it beats the strong production-default Fisher-weighted baseline by 17.9% served KL at 27B (two disjoint corpora) and a weaker output-MSE baseline by 38–40% at the 4B knee. The apparent additive "overshoot" that motivated the cross-layer apparatus is, at these bit-widths, largely a bf16 measurement floor; the genuine residual is small and is absorbed by selection. And the frontier's shape is rate–distortion geometry: convex, with sharp knees living in the format menu, not the sensitivity distribution. We have been explicit that the headline is a distributional-fidelity win, not a perplexity win, and that a weight-only cost is not a complete W4A4 model — the accounting a faithful method paper owes its reader.

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